XBATTERY

BharatBMS for Energy Storage

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Xbattery BharatBMS-ESS

Xbattery BharatBMS-ESS is a compact, high-performance BMS engineered for next-generation Energy Storage Systems. Supporting up to 42S LFP and NMC packs, it ensures precise monitoring, smart balancing, and multi-level protection for reliable and scalable energy management.

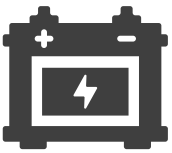
Applications

Backup & UPS Systems

Uninterrupted, stable energy support.

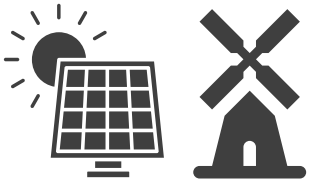
Telecom Infrastructure

Reliable backup for towers.



Industrial Battery Packs

High-performance power systems.



Renewable Energy Storage

Solar and wind power systems.

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Core Features

Intelligent Cell Monitoring

Supports up to 42S cells with ±2 mV precision and passive balancing.

Communication & Control

CAN FD, UART, SPI, and Ethernet for real-time data and remote monitoring.

Safety First

ISO 26262-compliant protection with fault diagnostics for voltage, current, and temperature.

Thermal Management

Real-time temperature sensing and thermal runaway detection for efficient heat control.

Scalable Architecture

Modular design allows flexible series and parallel configurations.

SOC/SOH Algorithms

Accurate State-of-Charge and State-of-Health estimation.

Smart Diagnostics

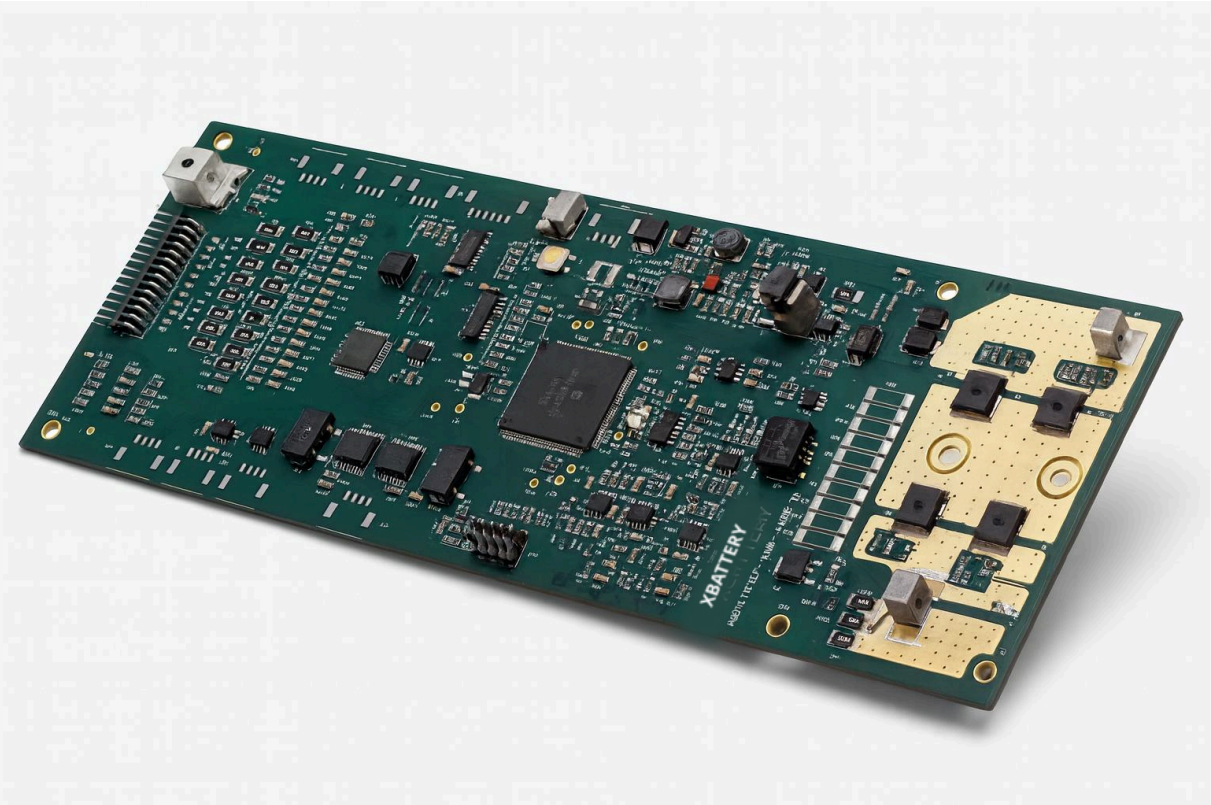
Real-time visualization and lifecycle analytics for optimal performance.

Optimized Software

High-efficiency C framework for real-time applications.

Customizable API

Open API for seamless integration and control.



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ESS BMS Offerings

01

Xbattery BharatBMS-ESS-48V

Modular smart BMS for UPS, telecom, and small ESS supports up to 16S for compact, reliable energy storage.

02

Xbattery BharatBMS-ESS-72V

High-efficiency BMS for industrial and medium-scale ESS, supports up to 28S for safe and efficient power management.

03

Xbattery BharatBMS-ESS-110V

Advanced BMS for large ESS and grid storage, supports up to 42S for high-voltage, high-performance energy systems.

Product Specifications

Specification	ESS 48V	ESS 72V	ESS 110V
Cell Configuration	Up to 16S	Up to 28S	Up to 42S
Nominal Pack Voltage Range	51.2V	72.0V	110.0V
Max Pack Voltage Range	58.4V	100.8V	151.2V
Min Pack Voltage Range	40.0V	70.0V	105.0V
Typical Applications	UPS, Telecom, Small ESS	Industrial ESS	Large ESS
Battery Capacity Range	5kWh	25kWh	38kWh

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Tech Specifications

General & Electrical Specifications	Parameters
Input Power & Communication Power Requirements	9V–60V input, 12V typical; CAN-powered or separate DC
Cell Voltage & Balancing	
Cell Voltage Measurement Range	1.5V to 4.2V
Measurement Accuracy & Resolution	±5mV; 1mV resolution
Balancing Topology (active or passive/ dissipative)	Passive
Balancing Current	~30 to 100mA (depending on configuration)
Balance Enable/Threshold Voltages	Typically enabled >3.45V
Temperature Sensing & Thermal Management	
Number & Type of Temperature Sensors Supported	4–8 NTCs (10kΩ typical)
Sensor Range & Accuracy	±1°C
Thermal Protection Thresholds and cut-offs	-20°C to +60°C (pack operation); <0°C or >60°C triggers cutoff
Current Handling	
Continuous Charge Current Limit	100A (based on 100Ah pack, 1C)
Peak Charge & Discharge Currents (duration and recovery conditions)	150A for <10s
Continuous Discharge and Peak Discharge Limits	100A/150A (<10s)
Internal Circuit Resistance	<5mΩ (BMU + cabling)
Standards & Certifications	
Safety Standards	AIS-038 Rev 2, IEC 62133-2
EMS/EMC compliance, regulatory notes	BIS applicable, CISPR 25 in roadmap
Performance & Lifecycle	
Efficiency	>98%
Power Consumption (active/idle/ sleep current)	200mW/50mW/<5mW
Cycle Life Expectations (if gauging capacity)	2000+ cycles (depends on cell)

Mechanical & Environmental	Parameters
Dimensions & Weight	Custom board, ~100mm × 70mm (est.)
Mounting Form Factor/Connector Types	Configurable
Operating & Storage Temperature Range	-20°C to +60°C ; -40°C to +70°C
Humidity/Environmental Classifications (indoor/outdoor, shock & vibration specs)	10–90% RH, non-condensing
Ingress/IP rating (if enclosure provided)	IP20–IP65 depending on enclosure
Software, Diagnostics & Tools	
Configuration Software Tools	PC-based GUI or CLI over UART/CAN
Event/Error Logging & RTC capabilities	Timestamped fault log (non-volatile) ; Yes (battery-backed optional)
Diagnostics & Self-Test features	Self-test on boot, watchdog, CAN alive
Versioning & Firmware upgrade method	UART or CAN bootloader supported
Protections & Safety	
Over-voltage Protection (per cell and pack)	Cell: >3.65V; Pack: >58.8V
Under-voltage Protection (per cell and pack)	Cell: <2.5V; Pack: <36V
Over & Under-temperature cut-offs	Charge: 0°C to 45°C; Discharge: -20°C to 60°C
Leakage/Ground fault detection	Optional (RCD/GFD integration)
Protection switching topology	MOSFET/Contactor based
Precharge circuit or inrush current limiting	Yes (resistive soft start/inrush limiter)
Communication Interfaces	
Supported Protocols	CAN 2.0B (500kbps), UART (optional), BLE (ESP8266)
Baud Rates & CAN Speeds	CAN: 125k to 1 Mbps; UART: 9600–115200
General-purpose I/O	6 (Analog Inputs)
Host Interfaces	CAN or UART to host/ controller/charger